Exploratory Data Analysis – School Dataset

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Project Overview

This project involves a comprehensive Exploratory Data Analysis (EDA) and predictive modeling of a school dataset containing 5,200 records. The primary goal is to understand the factors affecting student performance across Math, Language, and Science scores, while also evaluating school infrastructure and teacher-related metrics.

Dataset Description

The dataset consists of 5,200 rows and 17 columns, including:

- School Information: School ID, District, Block, and Urban/Rural classification.
- Academic Performance: Percentage scores in Math, Language, and Science.
- Staff Metrics: Teacher-Student ratio, average teacher experience, and training sessions attended.
- Infrastructure: Availability of electricity, internet, functional toilets, and libraries.
- Demographics: Percentage of marginalized students, home internet access, and parent literacy rates.

Workflow

1. Data Understanding & Profiling

- Utilized Pandas, NumPy, Matplotlib, Seaborn, and SciPy for analysis.
- Conducted initial data inspection to identify categorical variables and numerical distributions.
- Analyzed data spread across different Districts and Blocks.

2. Data Cleaning & Preprocessing

- Handling Missing Values: Identified and addressed missing data in Math scores, Teacher-Student ratios, Internet availability, and Parent Literacy rates.
- Imputation: Used median imputation for numerical gaps to maintain statistical integrity.
- Outlier Detection: Applied the Interquartile Range (IQR) method to identify anomalies in performance and infrastructure metrics.

3. Exploratory Analysis

- Univariate Analysis: Visualized distributions of core subject scores.

- Infrastructure Assessment: Evaluated the correlation between physical facilities and academic outcomes.
- Bivariate Analysis: Examined the impact of teacher experience and student-teacher ratios on performance.

4. Predictive Modeling

- Implemented a Linear Regression model to predict Math scores based on infrastructure and demographic features.
- Evaluated performance using R-squared (R^2) metrics and visualized Actual vs. Predicted results.

Technologies Used

- Python
- Pandas & NumPy
- Matplotlib & Seaborn
- Scikit-learn
- Jupyter Notebook